SEFI Canonical Mathematics

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1 Core Objects

Let $(\mathcal{M}, g_{\mu\nu})$ be a smooth Lorentzian manifold. SEFI introduces:

- Worldline field $x^\mu(\tau)$
- Tension scalar field $T(x)$
- Torsion drift tensor $\Theta_{\mu\nu}(x)$
- Coherence vector field $\Gamma_\mu(x)$
- Curvature-knot scalar $K(x)$

The standard curvature tensors are:

$$R^\rho{}_{\sigma\mu\nu}, \quad R_{\mu\nu}, \quad R = g^{\mu\nu} R_{\mu\nu}.$$

2 Continuity Invariant

Identity continuity along a worldline is defined by:

$$I[x^\mu] = \int_{\tau_0}^{\tau_1} \sqrt{g_{\mu\nu}(x) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}}\, d\tau,$$

with the SEFI requirement:

$$\delta I = 0 \quad \text{under tension-projection transformations.}$$

3 Warp-Metric Scalar

Define the SEFI warp-metric scalar:

$$W = R + \alpha T^2 + \beta \Theta_{\mu\nu}\Theta^{\mu\nu} + \gamma \Gamma_\mu \Gamma^\mu - \delta K^2,$$

where $\alpha, \beta, \gamma, \delta$ are SEFI coupling constants.

4 SEFI Lagrangian

The canonical SEFI Lagrangian density is:

$$\mathcal{L}_{\text{SEFI}} = \frac{1}{2\kappa}W + \frac{1}{2}\nabla_\mu T \nabla^\mu T + \frac{1}{4}\nabla_\rho \Theta_{\mu\nu}\nabla^\rho \Theta^{\mu\nu} + \frac{1}{2}\nabla_\mu \Gamma_\nu \nabla^\mu \Gamma^\nu - V(T, \Theta_{\mu\nu}, \Gamma_\mu, K),$$

where V is a potential encoding curvature-knot formation and stability:

$$V = \lambda_T T^4 + \lambda_\Theta (\Theta_{\mu\nu}\Theta^{\mu\nu})^2 + \lambda_\Gamma (\Gamma_\mu \Gamma^\mu)^2 + \lambda_K (K^2 - K_{\text{crit}}^2)^2.$$

The action is:

$$S_{\text{SEFI}} = \int_{\mathcal{M}} d^4x \, \sqrt{-g}\, \mathcal{L}_{\text{SEFI}}.$$

5 Field Equations

5.1 Metric Field Equation

Variation with respect to $g_{\mu\nu}$ gives:

$$\frac{1}{2\kappa}G_{\mu\nu} = T_{\mu\nu}^{(\text{SEFI})},$$

where $G_{\mu\nu}$ is the Einstein tensor and $T_{\mu\nu}^{(\text{SEFI})}$ is the SEFI stress-energy tensor derived from $\mathcal{L}_{\text{SEFI}}$:

$$T_{\mu\nu}^{(\text{SEFI})} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{SEFI}}}{\delta g^{\mu\nu}}.$$

5.2 Tension Field Equation

Variation with respect to T:

$$\nabla_\mu \nabla^\mu T + \alpha T + \frac{\partial V}{\partial T} = 0.$$

5.3 Torsion Drift Equation

Variation with respect to $\Theta_{\mu\nu}$:

$$\nabla_\rho \nabla^\rho \Theta_{\mu\nu} + \beta \Theta_{\mu\nu} + \frac{\partial V}{\partial \Theta^{\mu\nu}} = 0.$$

5.4 Coherence Field Equation

Variation with respect to Γ_μ:

$$\nabla_\nu \nabla^\nu \Gamma_\mu + \gamma \Gamma_\mu + \frac{\partial V}{\partial \Gamma^\mu} = 0.$$

5.5 Curvature-Knot Equation

Variation with respect to K:

$$-2\delta K + \frac{\partial V}{\partial K} = 0.$$

6 Warp-Metric Solutions

Warp-metric configurations satisfy:

$$W = \text{const},$$

with stability condition:

$$\frac{\delta^2 S_{\text{SEFI}}}{\delta \phi^2} > 0 \quad \text{for all SEFI fields } \phi \in \{T, \Theta_{\mu\nu}, \Gamma_\mu, K\}.$$

FTL-like warp configurations are defined as solutions where effective geodesic separation changes without violating global continuity:

$$I[x^\mu] = \text{invariant}, \quad \text{while } g_{\mu\nu} \to \tilde{g}_{\mu\nu} \text{ via SEFI warp.}$$

7 Quantization Sketch

Promote SEFI fields to operators:

$$T(x) \to \hat{T}(x), \quad \Theta_{\mu\nu}(x) \to \hat{\Theta}_{\mu\nu}(x), \quad \Gamma_\mu(x) \to \hat{\Gamma}_\mu(x), \quad K(x) \to \hat{K}(x),$$

with canonical commutation relations:

$$[\hat{T}(x), \hat{\Pi}_T(y)] = i\delta^{(4)}(x - y),$$

and similarly for $\Theta_{\mu\nu}, \Gamma_\mu, K$.
The SEFI Hamiltonian density is:

$$\mathcal{H}_{\text{SEFI}} = \sum_\phi \hat{\Pi}_\phi \dot{\phi} - \mathcal{L}_{\text{SEFI}},$$

where ϕ runs over all SEFI fields and $\hat{\Pi}_\phi$ are the conjugate momenta.
Physical states satisfy a SEFI constraint:

$$\hat{I}\,|\Psi\rangle = I_0\,|\Psi\rangle,$$

enforcing identity continuity at the quantum level.

8 Biological Manifold Mapping

A biological system is modeled as a submanifold $\mathcal{B} \subset \mathcal{M}$ with induced metric h_{ab}. SEFI fields restricted to $\mathcal{B}$:

$$T|_{\mathcal{B}}, \quad \Theta_{ab}|_{\mathcal{B}}, \quad \Gamma_a|_{\mathcal{B}}, \quad K|_{\mathcal{B}},$$

encode:

- Development: $\partial_t h_{ab} = U_{ab}(T, \Theta, \Gamma)$
- Metabolism: currents $J^a = \nabla_b T^{ab}$
- Immune correction: $C^a = -\nabla_b K^{ab}$
- Cancer: $K > K_{\text{crit}}$ on $\mathcal{B}$
- Healing: flow toward minima of V on $\mathcal{B}$

9 Consciousness Manifold

Consciousness is modeled as coherence across layered manifolds $\mathcal{M}_i$ with shared continuity invariant:

$$I_{\text{total}} = \sum_i I[\mathcal{M}_i],$$

subject to:

$$\delta I_{\text{total}} = 0 \quad \text{under SEFI warp and projection.}$$